Cross-Subject EEG Mental Fatigue Detection: Domain and Test-Time Adaptation

Bachelor Thesis Proposal

1 Problem

EEG-based fatigue detection models often perform well when trained and tested on the same subject but degrade when applied to unseen individuals. Cross-subject experiments typically show a performance drop of 15–30 percentage points.
The goal of the thesis is to evaluate how domain adaptation and test-time adaptation methods can improve cross-subject fatigue detection while minimizing the calibration effort required for new users.

2 Existing Work

Classical machine learning models using hand-crafted features such as band power and classifiers like logistic regression or SVM achieve **60–70%** accuracy in leave-one-subject-out (LOSO) evaluation [1, 2].

Deep learning models such as CNNs or EEGNet learn spatial–temporal representations automatically and generally reach **65–80%** cross-subject accuracy [3, 4].

Domain adaptation methods reduce distribution mismatch between subjects using techniques such as Transfer Component Analysis (TCA), Domain-Adversarial Neural Networks (DANN), and multi-source domain adaptation. These approaches report **70–90%** accuracy depending on dataset complexity [5, 6, 7].

Cross-Subject Studies

An overview of cross-subject studies using advanced domain adaptation strategies:

Study	Method	Dataset	Evaluation	Accuracy
Liu et al. (2020)	TCA / MIDA domain adaptation	Driving fatigue EEG	LOSO	68–73%
Zhao et al. (2021)	Multi-source domain adaptation	Fatigue EEG dataset	Cross-subject	∼85.9%
Chen et al. (2023)	FLDANN adversarial network	SEED-VIG	Cross-subject	84.1%
Huang et al. (2025)	Multi-level domain adaptation	SEED-VIG	Cross-subject	94.2%
Paulo et al. (2021)	CNN spatiotemporal encoding	Drowsiness EEG	LOSO	∼75.9%

3 Research Question

How do domain adaptation and test-time adaptation methods affect cross-subject generalization performance in EEG-based mental fatigue detection under different calibration settings, from no calibration (zero-shot) to unsupervised and few-shot adaptation?

Sub-questions

- What is the performance gap between within-subject and cross-subject EEG-based mental fatigue detection?
- To what extent can simple domain adaptation methods (e.g., Euclidean Alignment, Adaptive Batch Normalization) improve cross-subject generalization performance?
- Can test-time adaptation methods improve cross-subject performance without requiring labelled calibration data?
- How does model performance change across different calibration scenarios, ranging from no calibration to few-shot labelled adaptation?

4 Research Gap

Most domain adaptation methods perform **offline adaptation**, where target-domain data are used during training and the model remains fixed during inference [5, 6, 7]. Furthermore, many studies focus on complex deep domain adaptation architectures, while the effectiveness of simpler and more lightweight adaptation techniques has not been systematically studied for EEG-based mental fatigue detection.

There is also limited research on **test-time or online adaptation** for EEG fatigue detection. Similar adaptation techniques have shown strong performance in other EEG domains such as motor-imagery BCIs and sleep staging, where normalization-based adaptation and entropy-minimization methods enable deployment without calibration [8, 9]. However, these approaches have not yet been systematically evaluated for mental fatigue detection.

Also, analysis of the **accuracy–calibration trade-off** between zero-calibration models, unsupervised adaptation, and few-shot labelled calibration remains largely under-investigated.

5 Proposed Methodology

Experiments will use a public multi-subject dataset such as SEED-VIG. Evaluation will follow Leave-One-Subject-Out (LOSO), where in each fold the model is trained on all subjects except one and evaluated on the held-out subject.

Baseline models

- **Classical baseline**: band-power features (e.g., theta, alpha, beta power) with logistic regression or SVM.
- **Deep baseline**: EEGNet or a compact CNN trained on the source subjects in the LOSO training set.

Adaptation methods

- **Euclidean Alignment (EA)**: alignment of covariance matrices between source and target EEG distributions prior to classification.
- **Adaptive Batch Normalization (AdaBN)**: updating batch normalization statistics using unlabeled target-subject data.

- **TENT (Test-Time Entropy Minimization)** [8]: adapting model parameters during inference by minimizing prediction entropy on unlabeled target data.
- **Pseudo-label self-training**: high-confidence predictions are used as pseudo-labels for incremental model updates.

Few-shot calibration

- Fine-tuning the classification head using a small labeled calibration set (e.g., 10–100 EEG windows from the target subject) while keeping early feature extraction layers frozen.

To simulate real-world deployment, EEG windows from the target subject will be processed sequentially, allowing test-time adaptation methods to update the model during inference.

Performance will be evaluated using accuracy and balanced accuracy under LOSO evaluation. Adaptation methods will be compared based on the amount of calibration required (none, unlabeled target data, or small labeled calibration sets) in order to analyze the trade-off between predictive accuracy and calibration effort.

6 Backup Plan

If test-time adaptation is unstable or not effective, the project will focus on comparison of cross-subject domain adaptation methods and calibration strategies, including Euclidean alignment, AdaBN, and few-shot fine-tuning.

7 Additional Work

If time allows:

- Testing CNN–LSTM hybrid architectures
- Interpreting model predictions (e.g., feature importance or saliency maps) to analyze how EEG features used by the model change after adaptation
- A demo showing how accuracy changes with online adaptation

8 Timetable

Task	Estimated Hours
Literature review	40
Dataset preparation and preprocessing	40
Baseline model implementation	60
Domain adaptation experiments	80
Test-time adaptation experiments	60
Evaluation and analysis	30
Writing thesis report	40

9 Expected Outcomes

- Quantify cross-subject performance gaps for EEG fatigue detection
- Compare simple domain adaptation methods
- Evaluate test-time adaptation methods
- Analyze the trade-off between predictive accuracy and calibration effort

References

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